

Sumin Kang

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My research develops **provable-quality approximations and decomposition algorithms** for **large-scale stochastic mixed-integer optimization**, motivated by **logistics networks** and **critical infrastructure resilience**.

Academic Appointments

Postdoctoral Researcher (PI: Dr. Ward Romeijnnders) Sep 2025–present

Department of Operations, Faculty of Economics and Business
University of Groningen, The Netherlands

- Develop converging algorithms and approximations with error bounds for stochastic mixed-integer programs.
- Design novel cutting planes and computational frameworks; released an open-source package. [\[GitHub\]](#)

Education

Ph.D., Industrial and Systems Engineering Jan 2021–Aug 2025

Grado Department of Industrial and Systems Engineering (ISE)
Virginia Tech, Blacksburg, VA, USA

- **Dissertation:** Algorithms for Distributionally Risk-Receptive and Robust Stochastic Integer Programs and Interdiction Problems (Advisor: Dr. Manish Bansal) [\[Link\]](#)

M.S., Logistics Sep 2015–Jun 2017

Department of Air Transport, Transportation, and Logistics
Korea Aerospace University, South Korea

B.S., Logistics Mar 2012–Jun 2015

Department of Air Transport, Transportation, and Logistics
Korea Aerospace University, South Korea

Research Interests

Methodologies: Stochastic mixed-integer programming; distributionally robust optimization; decomposition methods; stochastic modeling.

Applications: Logistics networks; network interdiction (attacker–defender models); power-system restoration and operations.

Publications & Manuscripts

Journal Articles

1. **S. Kang** and M. Bansal, “Distributionally risk-receptive and robust multistage stochastic integer programs and interdiction models,” *Mathematical Programming*, 2025. [\[Link\]](#)
2. **S. Kang** and M. Bansal, “Distributionally risk-receptive and risk-averse network interdiction problems with general ambiguity set,” *Networks* 81(1), 3-22, 2022. [\[Link\]](#)

Manuscripts in Revision

3. W. Romeijnnders, N. van der Laan, **S. Kang**, “Benders decomposition with scaled cuts for multistage stochastic mixed-integer programs,” *SIAM Journal on Optimization* (*Revision submitted; Round 2*). [\[GitHub\]](#)

4. **S. Kang** and M. Bansal, “Bi-parameterized two-stage stochastic min-max and min-min mixed integer programs,” *INFORMS Journal on Computing (In revision; Round 1)*. [\[Link\]](#).

Working Papers

5. **S. Kang** and W. Romeijnders, “Partial Convexifications for two-stage stochastic mixed-integer programs,” Working paper (target submission: mid 2026).
6. **S. Kang**, M. Bansal, S.C. Sarin, and H. Nagarajan, “An integrated approach for power transmission system restoration,” Working paper (target submission: early 2026).
7. **S. Kang** and M. Bansal, “A decomposition algorithm for robust mixed-integer min-max problems,” In preparation.

Other Publications (prior to doctoral studies)

8. **S. Kang**, M. Kim, and J. Chae, “A closed loop based facility layout design using a cuckoo search algorithm,” *Expert Systems with Applications* 93, 322-335, 2018.
9. **S. Kang** and J. Chae, “Harmony search for the layout design of an unequal area facility,” *Expert Systems with Applications* 79, 269-281, 2017.
10. **S. Kang** and J. Chae, “A harmony search based algorithm for facility layout design with mixed type departments,” *Korean Journal of Logistics* 23.4, 1-15, 2015.

Research Experience

Visiting Research Intern

Jan 2025–Jun 2025

Los Alamos National Laboratory, NM, USA (Collaborated with Dr. Harsha Nagarajan)

- Tackled a power system restoration planning problem, integrating logistical planning and power system operational decisions.
- Developed valid inequalities and a column generation approach for restoration crew scheduling.

Research Fellow

Dec 2019–Dec 2020

Korea Aerospace University, South Korea

- Designed scalable algorithms and discrete-event simulations in industry-collaborated projects.

Teaching Experience

Instructor, Virginia Tech

Summer 2024

- ISE 3424 Discrete-Event Simulation (Undergraduate; enrollment: 26; overall rating: 5.6/6.0)
 - Designed and delivered lectures, assignments, and exams.
 - Developed hands-on labs and projects-based assignments using simulation software (Simio/@Risk); covered input modeling and statistical analysis for simulation output.

Teaching Assistant, Virginia Tech

2021–2024

- **Optimization / Operations Research:** ISE 5104 (Grad OR), ISE 5405 (Optimization I), ISE 2404 (Deterministic OR I).
- **Probability / Statistics:** ISE 5034 (Mathematical Probability & Statistics).
- **Simulation:** ISE 3424 (Discrete-Event Simulation).

Teaching Assistant, Korea Aerospace University

2015–2016

- AT 3202 Operations Research I; AT 4352 Analysis of Logistics System.

Honors and Awards

- Travel Award, Mixed Integer Programming (MIP) Workshop Student Poster Competition
Granted for selected presenters. Received in 2022, 2023, and 2024.
- Sanghani Center Student Spotlight, March 2024, [\[Link\]](#)
- Travel Award, The Sanghani Center for Artificial Intelligence and Data Analytics, 2023
- Outstanding Paper Award, Korea Logistics Society Spring Conference, 2017
- National Science and Engineering Undergraduate Scholarship, Korea Student Aid Foundation, 2014 and 2015
- Winner, the Undergraduate Capstone Design Competition, Korea Aerospace University, 2014
- Dean's Scholarship for Academic Excellence (1st ranked), Korea Aerospace University, 2012–2015

Presentations

Conference Presentations (2021–present)

- *Distributionally ambiguous multistage stochastic integer and disjunctive programs: applications to sequential two-player interdiction games*. INFORMS Annual Meeting, Phoenix, AZ, 2023.
- *Solution approaches for distributionally robust mixed-integer programs*. INFORMS Annual Meeting, Indianapolis, IN, 2022.
- *Distributionally risk-receptive and risk-averse network interdiction problems with general ambiguity set*. INFORMS Annual Meeting, Indianapolis, IN, 2022.
- *Attacker-defender maximum coverage location problem with partial coverage and spatial demand*. INFORMS Conference on Security, Arlington, VA, 2022; INFORMS Annual Meeting (Virtual), 2021.

Posters

- *Algorithms for bi-parameterized two-stage stochastic integer programs*. MIP Workshop (Poster Competition), Lexington, KY, 2024.
- *Distributionally ambiguous multistage stochastic integer and disjunctive programs: applications to sequential two-player interdiction games*. MIP Workshop (Student Poster), Los Angeles, CA, 2023.
- *Distributionally risk-receptive and risk-averse network interdiction problems with general ambiguity set*. MIP Workshop (Student Poster), New Brunswick, NJ, 2022.

Professional Services

- **Refereeing:** INFORMS Journal on Computing; IIE Transactions; Networks; Optimization Letters; Journal of Optimization Theory and Applications; Journal of Global Optimization; the INFORMS Journal on Optimization conference.
- **Conference/Community Service:** (Session Chair) INFORMS Annual Meeting 2022; INFORMS Optimization Society Conference 2022; INFORMS Conference on Security 2022.
- **Memberships:** INFORMS; SIAM; INFORMS Optimization Society; INFORMS Computing Society.
- **Affiliations/Leadership:** Affiliated student, the Sanghani Center for Artificial Intelligence & Data Analytics–Virginia Tech; Founder, the programming/research club at Korea Aerospace University

Computing Skills

- **Programming Languages:** Python, Java, Julia
- **Optimization:** Gurobi, CPLEX, JuMP, AMPL
- **Simulation:** Simio, @Risk, AutoMod, ExtendSim
- **Tools:** Git, HPC